Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: AdventureWorks Sales Analytics Dashboard& MySQL and connecting with any other sources into Power BI

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Experiment No.: 1

Importing AdventureWorks Sales Analytics Dashboard from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Github Link :

Problem Statement

The AdventureWorks dataset contains information related to sales transactions, customers, products and territories. The objective of this project is to use **Power BI** to transform the raw sales data into an interactive dashboard that helps analyze sales performance, order trends, customer distribution and product performance.
The dashboard provides management with a simple visual view of important business KPIs and trends.

Step 1: Data Collection

The main objectives of this project are:

- To analyze overall sales performance.
- To calculate total sales and total orders.
- To understand customer distribution.
- To analyze sales by product.
- To identify sales trends over time.
- To compare sales across territories.
- To analyze customers based on gender.
- To create an interactive Power BI dashboard.
- To support data-driven business decisions.

Step 2: Product Data Management Using MySQL

Opened MySQL Workbench to manage the Product dataset used in the project.
Created a database for storing and managing product-related information.
Imported the Product dataset into the MySQL database.
Verified the product records and important fields such as:

- ProductKey
- ProductName
- ProductSKU
- ProductColor
- ProductCost
- ProductPrice
- ProductSize
- ProductStyle

- ProductSubcategoryKey

Checked the product data to ensure that the required fields were available for analysis.

Used the MySQL database as a data source for the Power BI AdventureWorks Sales Dashboard.

Step 3: Data Import into Power BI

- Opened Power BI Desktop.
- Loaded the AdventureWorks Sales dataset.
- Imported the Sales table.
- Imported the Product table.
- Imported the Customer table.
- Reviewed important fields from the imported tables.
- Used the data to create measures and visualizations.

Built the AdventureWorks Sales Dashboard using the imported data

Step 4: Data Cleaning and Transformation

- Loaded the AdventureWorks dataset into Power BI.
- Checked the available tables and columns.
- Verified important fields such as ProductKey, CustomerKey, OrderDate, ProductName, ProductPrice, ProductColor and ProductStyle.
- Checked the data types of the required columns.
- Prepared the data for creating visualizations and dashboard analysis..

Step 5: Data Modeling

- Used the Sales, Product and Customer tables for analysis.
- Connected the tables using key fields such as ProductKey and CustomerKey.
- Used OrderDate for time-based analysis.
- Created a data model to analyze sales based on products, customers and dates.
- Used the model to build the AdventureWorks Sales Dashboard.

Step 6: DAX Measures

Created the following main measures:

- **Total Sales**
- **Total Orders**
- **Total Customers**
- For example:
- `Total Customers =`
- `DISTINCTCOUNT(Sales[CustomerKey])`

- Your dashboard results
- Total Sales: 24.91M

- Total Orders: 25K
- Total Customers: 17K

Step 7: Dashboard Development

- Developed the AdventureWorks Sales Dashboard using Power BI.
- Added Total Sales KPI – 24.91M.
- Added Total Orders KPI – 25K.
- Added Total Customers KPI – 17K.
- Created Total Sales by Month.
- Created Total Sales by Year.
- Created Top 5 Products by Total Sales.
- Created Total Orders by Product Name.
- Created Total Sales by Product Style.
- Created Total Sales by Territory.

Step 8: Interactive Filters and Slicers

- Added an Order Date slicer for date-based filtering.
- Added Product Name filtering.
- Added Gender filtering for customer analysis.
- Used slicers to make the dashboard interactive.
- Enabled users to analyze sales based on different products, dates and customer categories.

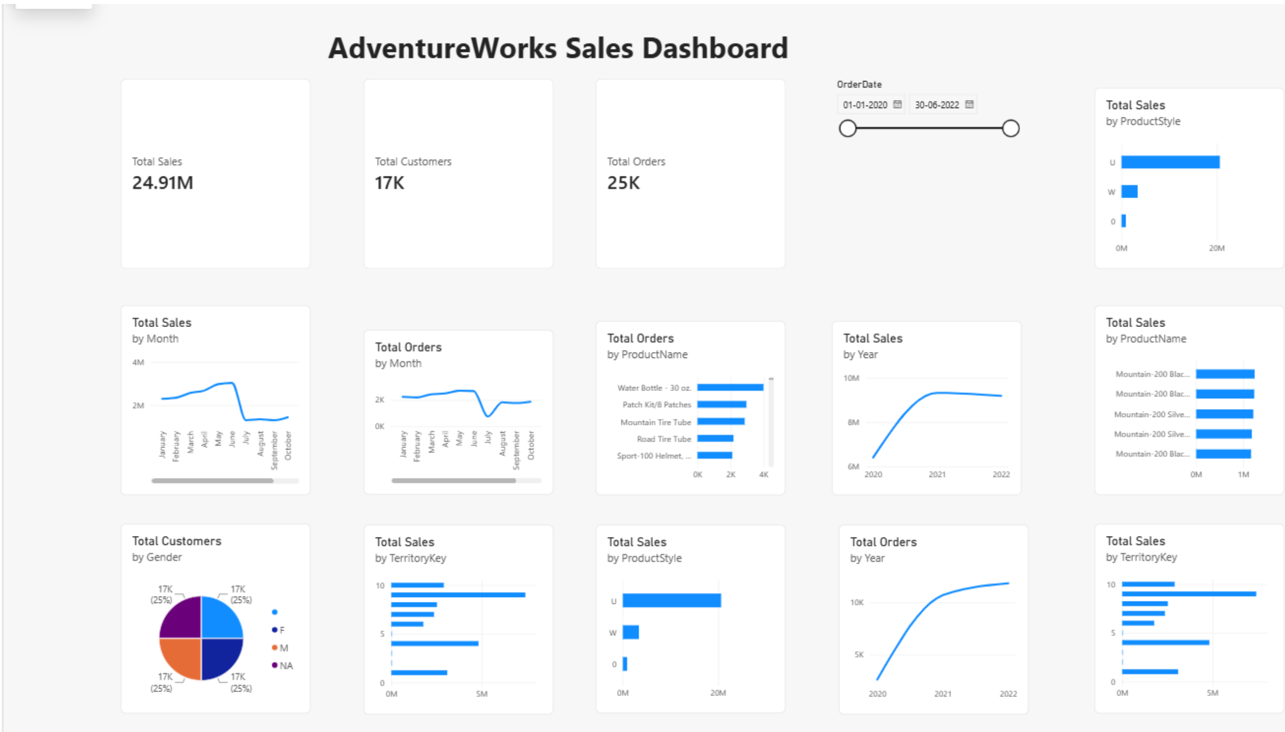
Step 9: Dashboard Formatting

- Added a clear AdventureWorks Sales Dashboard title.
- Arranged KPIs and charts in a structured layout.
- Adjusted chart sizes and positions.
- Added meaningful titles to the visuals.
- Applied consistent formatting for better readability.
- Kept the dashboard clean and uncluttered.

Step 10: Analysis and Insights

The completed dashboard was used to analyze:

- Overall sales performance.
- Total number of orders and customers.
- Monthly and yearly sales trends.
- Top-performing products.
- Sales performance by product style.
- Sales performance across territories.
- Customer gender distribution.
- Product and date-based sales performance



Step 11: Final Dashboard

- Completed the AdventureWorks Sales Dashboard using Power BI.
- Integrated the Sales, Product and Customer data.
- Created DAX measures for Total Sales, Total Orders and Total Customers.
- Added KPI cards, charts, slicers and filters.
- Created an interactive dashboard for analyzing sales performance.
- Used the dashboard to compare products, territories, dates and customers.

Conclusion

The AdventureWorks Sales Dashboard was successfully developed using Power BI.

The data was loaded, prepared and modeled to create meaningful sales visualizations. DAX measures were created for important KPIs such as Total Sales, Total Orders and Total Customers.

The final dashboard provides an interactive view of:

- Overall sales performance
- Total orders and customers
- Monthly and yearly sales trends
- Top-performing products
- Product style performance
- Territory-wise sales

- Customer gender analysis

Overall, the project demonstrates how Power BI can be used to transform business data into an interactive dashboard for better analysis and decision-making.

Learning Outcomes

Through this project, I learned:

1. **Data Collection and Integration**
   Learned how to load and work with the AdventureWorks Sales dataset in Power BI.
2. **Power BI Data Import**
   Learned how to import and work with the Sales, Product and Customer tables in Power BI.
3. **Data Cleaning and Preparation**
   Learned how to check columns, data types and important fields before creating visualizations.
4. **Data Modeling**
   Learned how to work with related tables using fields such as ProductKey, CustomerKey and OrderDate.
5. **DAX Measures**
   Learned how to create DAX measures for important KPIs such as Total Sales, Total Orders and Total Customers.
6. **Data Visualization**
   Learned how to create KPI cards, bar charts and line charts to represent sales data visually.
7. **Interactive Dashboard Development**
   Learned how to use slicers and filters to make the dashboard interactive.
8. **Sales Analysis**
   Learned how to analyze monthly and yearly sales trends, product performance and territory-wise sales.
9. **Customer Analysis**
   Learned how to analyze customer information, including total customers and gender-based analysis.
10. **Data-Driven Decision Making**
    Gained practical knowledge of how Power BI can transform raw business data into meaningful insights to support better decision-making